Lecture 11. Cox Proportional Hazards Regression (Part 1)

Semiparametric Model, Partial Likelihood, Wald/LR/Score Tests

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1 The Semiparametric Model

Data $(T_j, \delta_j, \mathbf{Z}_j)$, $j = 1, \ldots, n$, with $T_j = \min(X_j, C_j)$, $\delta_j = I(X_j \le C_j)$ and $\mathbf{Z}_j = (Z_{j1}, \ldots, Z_{jp})'$.

Definition 11.1: Proportional hazards (PH) model

$$h(t \mid \mathbf{Z}) = h_0(t)\, c(\boldsymbol{\beta}'\mathbf{Z}), \tag{8.1.1}$$

with h_0 the **baseline hazard** (at $\mathbf{Z} = \mathbf{0}$), completely unspecified; $c(\cdot) \ge 0$ a known function with $c(0) = 1$; $\boldsymbol{\beta} \in \mathbb{R}^p$. The usual choice $c(u) = e^u$ gives

$$h(t \mid \mathbf{Z}) = h_0(t) \exp(\boldsymbol{\beta}'\mathbf{Z}) = h_0(t) \exp\Big(\sum_{k=1}^{p} \beta_k Z_k\Big).$$

Semiparametric: the factor $c(\boldsymbol{\beta}'\mathbf{Z})$ is parametric with finite-dimensional $\boldsymbol{\beta}$; the baseline h_0 is nonparametric (infinite-dimensional).

1.1 Interpretation of $\boldsymbol{\beta}$

For two subjects with covariates $\mathbf{Z}_1, \mathbf{Z}_2$,

$$\frac{h(t \mid \mathbf{Z}_1)}{h(t \mid \mathbf{Z}_2)} = \frac{h_0(t) \exp(\boldsymbol{\beta}'\mathbf{Z}_1)}{h_0(t) \exp(\boldsymbol{\beta}'\mathbf{Z}_2)} = \exp\{\boldsymbol{\beta}'(\mathbf{Z}_1 - \mathbf{Z}_2)\}, \tag{8.1.3}$$

free of t: the **relative risk** (hazard ratio) of $\mathbf{Z}_1$ versus $\mathbf{Z}_2$.

1.2 Coding covariates

Example 11.2: larynx cancer, stage and age

90 males with larynx cancer. Entering stage $\in \{1,2,3,4\}$ as a single numeric covariate forces

$$\frac{h(t \mid \text{stage } k)}{h(t \mid \text{stage } 1)} = e^{(k-1)\beta}, \qquad k = 2,3,4,$$

i.e. log hazard ratios $\beta, 2\beta, 3\beta$ (an extra linearity assumption). Instead use dummies Z_1, Z_2, Z_3 ($= 1$ for stage 2, 3, 4; reference stage 1):

$$h(t \mid \mathbf{Z}) = h_0(t)\exp(\beta_1 Z_1 + \beta_2 Z_2 + \beta_3 Z_3 + \beta_4 \,\text{age}).$$

Interaction. The model assumes the same age effect at every stage. Add $Z_5 = Z_1 \cdot \text{age}$, $Z_6 = Z_2 \cdot \text{age}$, $Z_7 = Z_3 \cdot \text{age}$ and test $H_0 : \beta_5 = \beta_6 = \beta_7 = 0$.

Example 11.3: kidney transplant, race and gender

863 patients (432 white males, 92 black males, 280 white females, 59 black females).

- **Four groups** (reference: white female): Z_1, Z_2, Z_3 indicate black male, white male, black female.
- 2×2 **factorial**: $Z_1 = I(\text{male})$, $Z_2 = I(\text{black})$, $Z_3 = Z_1 Z_2$ (interaction).

2 Partial Likelihood

Motivation. h_0 is infinite-dimensional, so it is unclear that the usual parametric MLE theory applies. The partial likelihood estimates $\boldsymbol{\beta}$ separately from h_0.

2.1 Notation

Ordered distinct event times $t_1 < \cdots < t_D$, **no ties** ($d_i = 1$); $\mathbf{Z}_{(i)}$ the covariate of the subject failing at t_i (unique by the no-tie assumption); risk set

$$R(t_i) = \{j : T_j \geq t_i\}.$$

Censoring is **conditionally independent**: given $\mathbf{Z}_j$, event and censoring times are independent.

Definition 11.4: Partial likelihood

$$L(\boldsymbol{\beta}) = \prod_{i=1}^{D} \frac{\exp(\boldsymbol{\beta}'\mathbf{Z}_{(i)})}{\sum_{j \in R(t_i)} \exp(\boldsymbol{\beta}'\mathbf{Z}_j)}. \tag{8.3.1}$$

Heuristic. Each factor is a conditional probability:

$$P\{\text{subject with } \mathbf{Z}_{(i)} \text{ dies at } t_i \mid \text{one death at } t_i,\ R(t_i)\}$$
$$= \frac{h(t_i \mid \mathbf{Z}_{(i)})}{\sum_{j \in R(t_i)} h(t_i \mid \mathbf{Z}_j)} = \frac{h_0(t_i)\exp(\boldsymbol{\beta}'\mathbf{Z}_{(i)})}{\sum_{j \in R(t_i)} h_0(t_i)\exp(\boldsymbol{\beta}'\mathbf{Z}_j)} = \frac{\exp(\boldsymbol{\beta}'\mathbf{Z}_{(i)})}{\sum_{j \in R(t_i)} \exp(\boldsymbol{\beta}'\mathbf{Z}_j)}.$$

$h_0(t_i)$ cancels, so L does not involve h_0.

Example 11.5: toy data

Five subjects, all failing: $(t,z) = (1,3),(5,2),(9,4),(20,9),(22,4)$; $h_j(t) = h_0(t)\exp(\beta_1 z_j)$.

$$L(\beta_1) = \frac{e^{3\beta_1}}{e^{3\beta_1}+e^{2\beta_1}+e^{4\beta_1}+e^{9\beta_1}+e^{4\beta_1}} \cdot \frac{e^{2\beta_1}}{e^{2\beta_1}+e^{4\beta_1}+e^{9\beta_1}+e^{4\beta_1}} \cdot \frac{e^{4\beta_1}}{e^{4\beta_1}+e^{9\beta_1}+e^{4\beta_1}} \cdot \underbrace{\frac{e^{9\beta_1}}{e^{9\beta_1}+e^{4\beta_1}}}_{(A1)}.$$

(The last failure, alone in its risk set, contributes 1.)

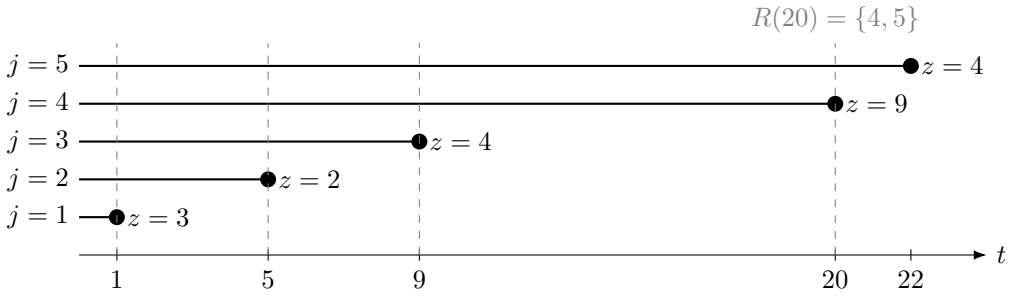


Figure 1: Toy data: a failure at t_i is compared with everyone still at risk.

Derivation: why factor (A1) is a conditional probability

For small Δ let $E_i = \{20 \le T_i < 20+\Delta\}$, $F_i = \{T_i \ge 20+\Delta\}$, $G_i = \{T_i \ge 20\}$. Given that subjects 4 and 5 survive to 20 and exactly one fails in $[20, 20+\Delta)$,

$$\begin{aligned} P(E_4, F_5 \mid G_4, G_5, \text{one failure}) &= \frac{P(E_4, F_5 \mid G_4, G_5)}{P(E_4, F_5 \mid G_4, G_5) + P(E_5, F_4 \mid G_4, G_5)} \\ &= \frac{\{P(E_4 \mid G_4)/\Delta\} P(F_5 \mid G_5)}{\{P(E_4 \mid G_4)/\Delta\} P(F_5 \mid G_5) + \{P(E_5 \mid G_5)/\Delta\} P(F_4 \mid G_4)}. \end{aligned} \tag{A3}$$

As $\Delta \to 0$, $P(E_i \mid G_i)/\Delta \to h_i(20)$ and $P(F_i \mid G_i) \to 1$, so (A3) $\to h_4(20)/\{h_4(20)+h_5(20)\} =$ (A1).

3 Estimation and Tests

Treat $L(\boldsymbol{\beta})$ as an ordinary likelihood.

Theorem 11.6: Maximum partial likelihood estimator

With $LL(\boldsymbol{\beta}) = \log L(\boldsymbol{\beta}) = \sum_{i=1}^{D} \left[\boldsymbol{\beta}'\mathbf{Z}_{(i)} - \log \sum_{j \in R(t_i)} \exp(\boldsymbol{\beta}'\mathbf{Z}_j)\right]$, $\hat{\boldsymbol{\beta}}$ solves $\mathbf{U}(\hat{\boldsymbol{\beta}}) = \mathbf{0}$, where

$$\mathbf{U}(\boldsymbol{\beta}) = \frac{\partial LL}{\partial \boldsymbol{\beta}} = \sum_{i=1}^{D} \Big[\mathbf{Z}_{(i)} - \frac{\sum_{j \in R(t_i)} \mathbf{Z}_j \exp(\boldsymbol{\beta}'\mathbf{Z}_j)}{\sum_{j \in R(t_i)} \exp(\boldsymbol{\beta}'\mathbf{Z}_j)}\Big],$$

solved numerically (Newton–Raphson). Then $\hat{\boldsymbol{\beta}} \approx N(\boldsymbol{\beta}_0,\, I^{-1}(\boldsymbol{\beta}_0))$ with

$$I(\boldsymbol{\beta}) = -\frac{\partial^2 LL}{\partial \boldsymbol{\beta}\, \partial \boldsymbol{\beta}'} = \sum_{i=1}^{D} \Big[\frac{\sum_{j \in R(t_i)} \mathbf{Z}_j^{\otimes 2} e^{\boldsymbol{\beta}'\mathbf{Z}_j}}{\sum_{j \in R(t_i)} e^{\boldsymbol{\beta}'\mathbf{Z}_j}} - \Big(\frac{\sum_{j \in R(t_i)} \mathbf{Z}_j e^{\boldsymbol{\beta}'\mathbf{Z}_j}}{\sum_{j \in R(t_i)} e^{\boldsymbol{\beta}'\mathbf{Z}_j}}\Big)^{\otimes 2}\Big], \tag{8.3.4}$$

$\mathbf{v}^{\otimes 2} = \mathbf{v}\mathbf{v}'$. (R: `coxph`.)

Definition 11.7: Three global tests of $H_0 : \beta = \beta_0$

Each is approximately χ^2_p under H_0:

1. Wald: $\chi^2_W = (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)' I(\hat{\boldsymbol{\beta}})(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)$;
2. partial likelihood ratio: $\chi^2_{LR} = 2\{LL(\hat{\boldsymbol{\beta}}) - LL(\boldsymbol{\beta}_0)\}$;
3. score: $\chi^2_{SC} = \mathbf{U}(\boldsymbol{\beta}_0)' I^{-1}(\boldsymbol{\beta}_0)\mathbf{U}(\boldsymbol{\beta}_0)$, from $\mathbf{U}(\boldsymbol{\beta}_0) \approx N(\mathbf{0}, I(\boldsymbol{\beta}_0))$.

4 A Single Binary Covariate

Example 11.8: breast cancer, new lymph-node classification

45 node-negative patients (old method); the new method classifies 9 positive, 36 negative. Fit $h(t \mid Z) = h_0(t)e^{\beta Z}$, $Z = I(\text{positive})$. Let Y_{0i}, Y_{1i} be the numbers at risk in the negative / positive group at t_i and $d_1 = \sum_i Z_{(i)}$ the deaths in the positive group. Then

$$LL(\beta) = \beta d_1 - \sum_{i=1}^{D} \log(Y_{0i} + Y_{1i}e^{\beta}), \qquad U(\beta) = d_1 - \sum_{i=1}^{D} \frac{Y_{1i}e^{\beta}}{Y_{0i} + Y_{1i}e^{\beta}},$$

$$I(\beta) = \sum_{i=1}^{D} \Big[\frac{Y_{1i}e^{\beta}}{Y_{0i} + Y_{1i}e^{\beta}} - \frac{Y_{1i}^2 e^{2\beta}}{(Y_{0i} + Y_{1i}e^{\beta})^2}\Big].$$

Score test is easiest by hand (no $\hat{\beta}$ needed). With $Y_i = Y_{0i} + Y_{1i}$ and $d_i = 1$,

$$U(0) = \sum_{i=1}^{D} \Big(d_{1i} - Y_{1i}\frac{d_i}{Y_i}\Big) = Z_1(\tau), \qquad I(0) = \sum_{i=1}^{D} \frac{Y_{1i}}{Y_i}\Big(1 - \frac{Y_{1i}}{Y_i}\Big) = \hat{\sigma}_{11},$$

so $\chi^2_{SC} = U(0)^2/I(0)$ is the **two-sample log-rank test** (Lecture 9).
Data (months; + censored). Negative: 19, 25, 30, 34, 37, 46, 47, 51, 56, 57, 61, 66, 67, 74, 78, 86, 122+, 123+, 130+, 130+, 133+, 134+, 136+, 141+, 143+, 148+, 151+, 152+, 153+, 154+, 156+, 162+, 164+, 165+, 182+, 189+. Positive: 22, 23, 38, 42, 73, 77, 89, 115, 144+.
Results: $\hat{\beta} = 0.9802$, relative risk $e^{0.9802} = 2.67$;

$$\chi^2_W = 5.08 \ (p = 0.02), \qquad \chi^2_{LR} = 4.45 \ (p = 0.04), \qquad \chi^2_{SC} = 5.49 \ (p = 0.02).$$

5 Selected exercises

Motivation. The log partial likelihood is concave, so the Newton–Raphson root is the maximiser; and the toy data make every formula concrete.

Exercise 11.1 (Concavity and invariance of the partial likelihood)

(a) Show that the ith term of $I(\boldsymbol{\beta})$ in (8.3.4) is the covariance matrix of $\mathbf{Z}_J$, where J is drawn from $R(t_i)$ with $P(J = j) \propto e^{\boldsymbol{\beta}'\mathbf{Z}_j}$. Conclude that LL is concave.
(b) Show that $L(\boldsymbol{\beta})$ is unchanged when every $\mathbf{Z}_j$ is replaced by $\mathbf{Z}_j + \mathbf{c}$ for a fixed vector $\mathbf{c}$. Why does the Cox model have no intercept?

Solution [Supplement].

(a) With $\pi_{ij} = e^{\boldsymbol{\beta}'\mathbf{Z}_j} / \sum_{l \in R(t_i)} e^{\boldsymbol{\beta}'\mathbf{Z}_l}$, the term is $\sum_j \pi_{ij} \mathbf{Z}_j^{\otimes 2} - \left(\sum_j \pi_{ij} \mathbf{Z}_j\right)^{\otimes 2} = \mathrm{Cov}(\mathbf{Z}_J) \succeq 0$. A sum of positive semidefinite matrices is positive semidefinite, so $-\partial^2 LL / \partial\boldsymbol{\beta}\partial\boldsymbol{\beta}' \succeq 0$.

(b) Each factor of (8.3.1) gains $e^{\boldsymbol{\beta}'\mathbf{c}}$ in numerator and denominator. An intercept β_0 multiplies h_0 by e^{β_0}, which is absorbed in the arbitrary h_0 and cancels from L. ■

Exercise 11.2 (Score test on the toy data)

For Example 11.5, compute $U(0)$, $I(0)$ and χ^2_{SC}.

Solution [Supplement]. At $\beta = 0$ each term of U is $z_{(i)} - \bar{z}_{R(t_i)}$ and each term of I is the (divide-by-$|R|$) variance of z over $R(t_i)$:

t_i	$R(t_i)$	$z_{(i)}$	$\bar{z}_R$	$z_{(i)} - \bar{z}_R$	$\mathrm{Var}_R(z)$
1	$\{3, 2, 4, 9, 4\}$	3	4.40	−1.40	5.84
5	$\{2, 4, 9, 4\}$	2	4.75	−2.75	6.69
9	$\{4, 9, 4\}$	4	5.67	−1.67	5.56
20	$\{9, 4\}$	9	6.50	2.50	6.25

$U(0) = -3.32$, $I(0) = 24.33$, $\chi^2_{SC} = 3.32^2/24.33 = 0.45$ ($p = 0.50$). (The time-22 term is 0.) ■